

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 1.4 TRANSPORT IN PLANTS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the driving question: How does a maize plant in a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves, and what happens to that water in the end? Think back to the two puzzling observations in Lesson 1: the red dye that travelled through a maize seedling from the Kisumu school garden, and the water droplets that formed inside the sealed polythene bag around a sweet potato cutting in afternoon sun. Answer every section in the blank space under each prompt. Use full sentences and scientific words from our lessons. Draw labelled diagrams where a prompt asks for one. Link each answer to the evidence we collected in class. Use your own thinking and what you observed.

Part 1: Why the dye appeared only in certain places

In Lesson 1, a maize seedling from the Kisumu school garden stood with its roots in red-dyed water for 30 minutes. When the stem was cut across, vivid red rings appeared only in specific regions, while the outer green tissue and the central pith stayed white.

- (a) Which tissue turned red? Explain how you know it is this tissue and not phloem.
- (b) Describe two features of this tissue that help water move upward through it.
- (c) Use the capillary tube demonstration to explain how adhesion and cohesion help water rise in narrow tubes.

Write your answer here:

Part 2: Getting water and minerals into the root

A farmer in the Rift Valley applies DAP fertiliser to her maize.

- (a) Explain how water enters a root hair cell, using what you learned about the cell sap and the soil water.**
- (b) Explain how phosphate ions from the DAP enter the root hair cell. Say why this process is different from the way water enters.**
- (c) Explain how the structure of a root hair cell suits its job.**
- (d) What can happen to maize or sukuma wiki if too much fertiliser is applied? Explain why.**
- (e) A maize plant in Mombasa has wilted in the afternoon heat but recovers after it is watered in the evening. Explain how the plant's cells recover.**

Write your answer here:

Part 3: Reaching the topmost leaves and carrying sugar

A maize plant can be about two metres tall.

- (a) Root pressure pushes water into the xylem. Is this push enough to raise water to the topmost leaves of the plant? Explain, and describe what else is involved.**
- (b) Name the tissue that carries sugars, state the process, and say where sugars travel from and to.**
- (c) In the ring-barking (girdling) demonstration, tissue swelled just above the cut but not below. Explain what this shows and why a girdled stem eventually dies even though water still reaches its leaves.**

Write your answer here:

Part 4: What happens to the water in the end

Think about the sweet potato cutting sealed in the polythene bag in afternoon sun.

(a) Where did the water droplets inside the bag come from? Trace the journey of that water from the soil to the bag.

(b) In our group experiments (Groups A to D) we changed stomata, wind, humidity and light. Using the results your class recorded and graphed, describe how each factor affected the amount of water lost, and explain the pattern using the concentration gradient of water vapour.

(c) Give three reasons why transpiration is useful to the maize plant rather than just a loss of water.

Write your answer here:

Part 5: Putting it all together

Write a full explanation that answers the driving question: How does a maize plant in a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves, and what happens to that water in the end?

Use a labelled diagram of a maize plant (roots, stem, leaf) with arrows for the movement of water, minerals and sugars. Then use your diagram in your written explanation. Finally, explain why transport is important to a plant. Add one piece of advice for a farmer in Kisumu based on what you learned.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of the transport explanation	Correctly explains osmosis, active transport, root pressure, transpiration pull, xylem and phloem roles with all key terms used accurately.	Explains most processes correctly with minor errors or one missing mechanism.	Explanation has major errors or confuses xylem and phloem, or osmosis and active transport.
Use of evidence from the lessons	Consistently links claims to the red-dye maize stem, the polythene bag, the girdling demonstration and the transpiration group results.	Refers to some of the evidence, but links are sometimes unclear.	Little or no reference to class evidence.
Explaining transpiration and gradients	Explains the effect of wind, humidity, light and blocked stomata using the water vapour concentration gradient, and gives three uses of transpiration.	Describes the effect of the factors, with a partial explanation of the gradient or fewer uses.	Only lists results without a mechanism.
Linking to the driving question and Kenyan context	Gives a coherent whole-journey explanation, a clear labelled diagram, and sensible advice relating to Kenyan farming.	Gives a mostly complete journey with a diagram and some context.	Explanation is fragmented, with a missing or unclear diagram and no application.
Communication	Writes clearly in logical order with correct scientific vocabulary.	Writing is mostly clear with some vocabulary slips.	Writing is hard to follow and vocabulary is mostly everyday.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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